Day - 1

Problem Framing - Process of analyzing a problem to isolate the individual elements that need to be addressed to solve it

We confirm that the non-ML approach first is optimized. If we dont have a non-ML based solution implemented then we try solving using heuristic
↳ short or quick method

Proxy labels → They substitute for labels that are not in the dataset.

Student Handout

Problem - 1

Verdict ML To stop fraudulent transfer

Justification Prediction needs to take place before money moves & Model can be trained to find patterns as we have 2 years of historical data

2 ← Checklist = define outcome, model output, success metrics

Task type or algorithm Binary Classification, checks fraud or not fraud

Prediction and inputs If a transaction is fraudulent or not from inputs like amount, time, device, location, payee

Success metric and accuracy Dataset is highly imbalanced & hence accuracy alone will not work, but here recall and precision are the most important, and the trade-off

between recall and precision.

actual fraud caught

flagged as fraud and are fraudulent

Simplest baseline to beat: Can make a heuristic formula / rules and then can manually classify or some prediction with assumption.

Risk or failure: False negative. as the model should not detect transaction which are not fraud as the predictive power can be lost.

Problem - 2

Verdict: Not ML

Justification: Only bill calculation, no historic data, model which is time consuming and expensive is not required no pattern to identify, Nothing to predict

Task type or Heuristic: Rule based calculation and mathematical formulas

Prediction and inputs: No predictions, only exact bill calculation required from inputs like units consumed, tariffs, rates, fixed charges and duty.

Success metric and Accuracy: The total of the bill should be correct, that's it

Baseline to beat — Heuristic / formulas / ~~equation~~

Risk or failure — The total can be wrong and at the end with costumer and others, it could create problems in asking or giving money

Problem - 3

Verdict — ML

Justification — Advice to be given and so prediction compulsory, a proper ~~forecast~~ forcast required and 5 years of historic data is given

Task type or alternate — Forecasting, for one ~~week~~ week it can go

Prediction and inputs — future prices from inputs like daily arrival quantities, rainfall records, daily prices from local market.

Success metric and accuracy — Forecasting error, measures how close the predicted prices is to the actual price, simple comparision.

Baseline to beat — Use recent observed price to forecast, methods like mean, naive approach or simple forcasting or seasonal approach.

Risk or failure — Inaccurate forecast lead to loss and errors will be there.

Problem - 4

Verdict: ML

Justification: Clustering or unsupervised learning needed, features like daily milk volume, payment histories, etc are are given ∴ we can find similarities and then can assign services to everyone

Task Type or alternate: Unsupervised learning

Prediction and inputs: No prediction, just grouping from inputs like fat percentage, daily milk volume, payment history and herd size

Success metric and accuracy: Quality of clusters, ability to discover hidden patterns

Baseline to beat: Heuristics, business rules manually group by features

Risk or failure: Poor grouping can lead to inappropriate services Can even discover meaningful patterns

Problem - 5

Verdict: No ML

Justification: Simple algo can also work and no unknown outcome or Prediction to be made

Task Type — Scheduling using constraints

Predicted and inputs — No prediction, only assign from inputs like faculty list with department, the room list, exam timetable and last years ~~~~ courses

Success metric — A valid course with no clashes, no invigilation in their own department and balanced assignment.

Baseline to beat — Manually constructed, rule based

R&R or Failures — Violates one or more constraints

~~Bonus~~ Bonus Problem

Verdict — ML

Justification — Prediction is required and also historical data is given of last 5 batches

Task type or alternate — Binary classification as dropout or not dropout and supervised learning

Predicted and inputs — Whether a first year student is likey to dropout or not can be predicted from inputs like attendance, internal marks, fee status, hostel records, home region and gender for the last 5 batches.